
 $^1\text{H}(^{35}\text{Cl}, 2\text{n})$ **2020Ia01**

2020Ia01: a 30-MeV/nucleon ^{35}Cl beam was produced from the K500 cyclotron at Texas A&M University and impinged on a liquid-nitrogen-cooled hydrogen gas target. The fully stripped reaction products recoiled into the Momentum Achromat Recoil Spectrometer (MARS), where they were spatially separated according to their charge-to-mass ratio. A 99.5% pure ^{34}Ar beam was implanted into aluminized Mylar tape and moved to a well-shielded location, where a 4π proportional gas counter was used for detecting positrons. A total of 1.7×10^8 decay positrons were recorded. Measured $E\beta$, $\beta(t)$. Deduced $T_{1/2}$ of ^{34}Ar .

2020Ia02: similar ^{34}Ar production setup at MARS. A 99.6% pure ^{34}Ar beam was implanted into aluminized Mylar tape and moved to the counting location, where a 1-mm-thick BC-404 scintillator was used to detect positrons and a HPGe detector was used to detect γ rays. Measured $E\beta$, $E\gamma$, $\beta(t)$, $\gamma(t)$, $\beta\gamma$ coin. A total of $2.6710(20) \times 10^8$ decay positrons were recorded.

 ^{34}Ar Levels

<u>E(level)</u>	<u>J^π</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0	0^+	0.84646 s 35	$T_{1/2}$: from $\beta(t)$ using two component fit to the decay of ^{34}Ar and ^{34}Cl (2020Ia01). The $T_{1/2}$ of ^{34}Cl was fixed to the world average of 1526.55 ms 44. The $T_{1/2}$ of ^{34}Ar with only statistical uncertainty is 0.84646 s 19.